

IGCSE • Cambridge (CIE) • Biology

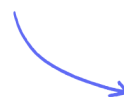
 50 mins  8 questions

Target Test

Created 30/05/26

Total Marks /50

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1 (a) This question is about enzymes.

Choose words from the list to complete the sentences.

Each word may be used once, more than once, or not at all.

amino acid	amylase	bacteria	biological
fatty acid	glucose	humidity	lipase
living	protease	salivary	temperature

All enzymes are proteins. Proteins are made of

An enzyme acts as acatalyst.

In order to work rapidly, enzymes need the correct and pH.

An example of an enzyme that works in the acidic conditions in the stomach is

Acidic conditions will kill many of the present in food.

Salivary stops working in acidic conditions.

Fats are broken down by

(7 marks)

(b) Match up the key terms to their definition.

Enzyme		Groove in enzyme where the reaction take place
Substrate		Molecules that bind to an enzyme
Catalyst		One enzyme for one substrate
Specificity		Any molecule that speeds up reactions
Active site		A protein that speeds up reactions

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..... (5 marks)

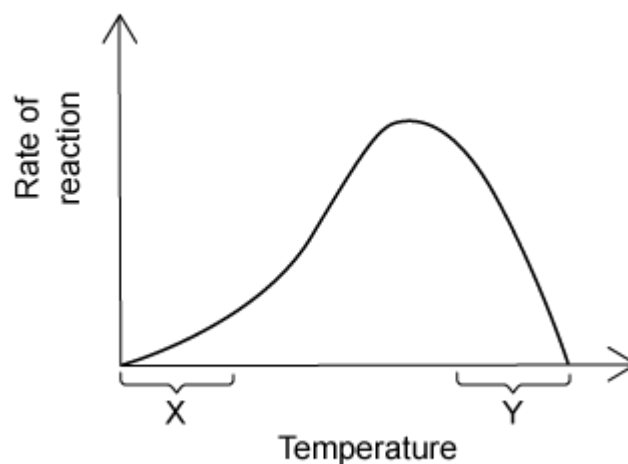
(c) **Extended**

Define what is meant by the term *enzyme-substrate complex*.

..... (1 mark)

2 Extended

Temperature affects the rate of all enzyme-controlled reactions, as shown in the graph below.



Which row correctly describes what is happening at the points on the graph?

	X	Y
A	the enzyme has been denatured	the rate of reaction is decreasing
B	the rate of reaction is increasing	the enzyme is becoming denatured
C	the reaction is taking less time	the rate of reaction is increasing
D	the rate of reaction is decreasing	the enzyme is becoming denatured

(1 mark)

3 Why are enzymes said to be specific?

- A.** Each enzyme can only work at one temperature.
- B.** Each enzyme has an active site with a complementary shape to a particular substrate.
- C.** The active site changes permanently after every reaction, so only one substrate can be used.
- D.** Each enzyme works best with substrates of the same size.

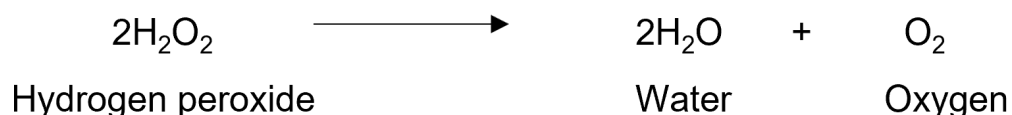
(1 mark)

4 Which statement describes a catalyst?

- A.** a substance that decreases the rate of a chemical reaction and is not changed by the reaction
- B.** a substance that decreases the rate of a chemical reaction and is changed by the reaction
- C.** a substance that increases the rate of a chemical reaction and is changed by the reaction
- D.** a substance that increases the rate of a chemical reaction and is not changed by the reaction

(1 mark)

5 (a) Catalase is an enzyme found in plant and animal cells. It catalyses the breakdown of hydrogen peroxide to form water and oxygen.



The oxygen produced forms a foam. You can measure the height of the foam to determine catalase activity.

A student investigated catalase activity in cooked and uncooked potato tissue.

The student:

Step 1	Cut two potato cylinders so that they were identical in shape and size
Step 2	Put 5 cm ³ of 3 % hydrogen peroxide solution into a test-tube labelled cooked potato
Step 3	Put 5 cm ³ of 3 % hydrogen peroxide solution into a test-tube labelled uncooked potato
Step 4	Put one of the potato cylinders into a beaker of hot water for five minutes
Step 5	Removed the potato cylinder from the hot water and put it into the test-tube labelled cooked potato
Step 6	Put the uncooked potato cylinder into the test-tube labelled uncooked potato
Step 7	Left the potato cylinders in the 3 % hydrogen peroxide solution for three minutes and then measured the height of the foam produced in each of the test-tubes.

Fig. 1 shows the test-tubes after three minutes.

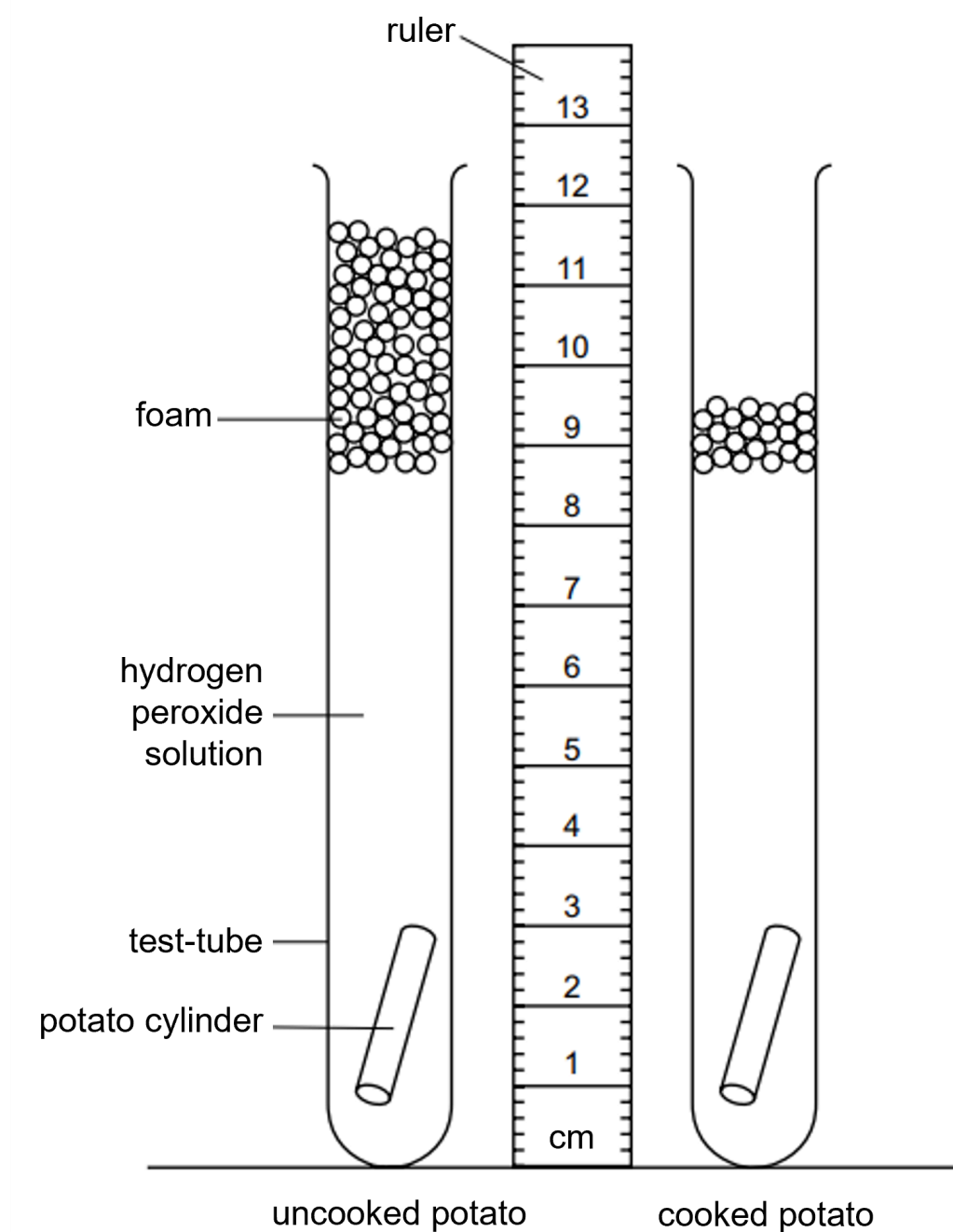


Fig. 1

(i)

Prepare a table to record your results.

Measure the height of the foam in each of the test-tubes in Fig. 1 Record these measurements in your table.

[3]

(ii)

Calculate the difference in the height of the foam produced by the cooked and uncooked potato after three minutes.

[1]

(iii)

State **one** conclusion for these results.

[1]

(5 marks)

(b) (i)

Identify the independent variable in this investigation.

[1]

(ii)

State why it is important that the two potato cylinders were identical in shape and size.

[1]

(iii)

State **two other** variables that were kept constant in this investigation.

[2]

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(4 marks)

(c) Identify **one** possible source of error in the method used in this investigation.

.....

(1 mark)

(d) A student stated that:

'Catalase activity is the same in all species of plants'

Plan an investigation to test this statement.

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..... (6 marks)

(e) Potatoes contain starch. Starch can be broken down into reducing sugars.

Describe the tests to identify starch and reducing sugars and give the results of the positive tests.

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..... (5 marks)

- 6 (a)** Catalase is an enzyme that catalyses the breakdown of hydrogen peroxide into water and oxygen.

The effect of temperature on catalase activity was investigated.

The results are shown in Fig. 1

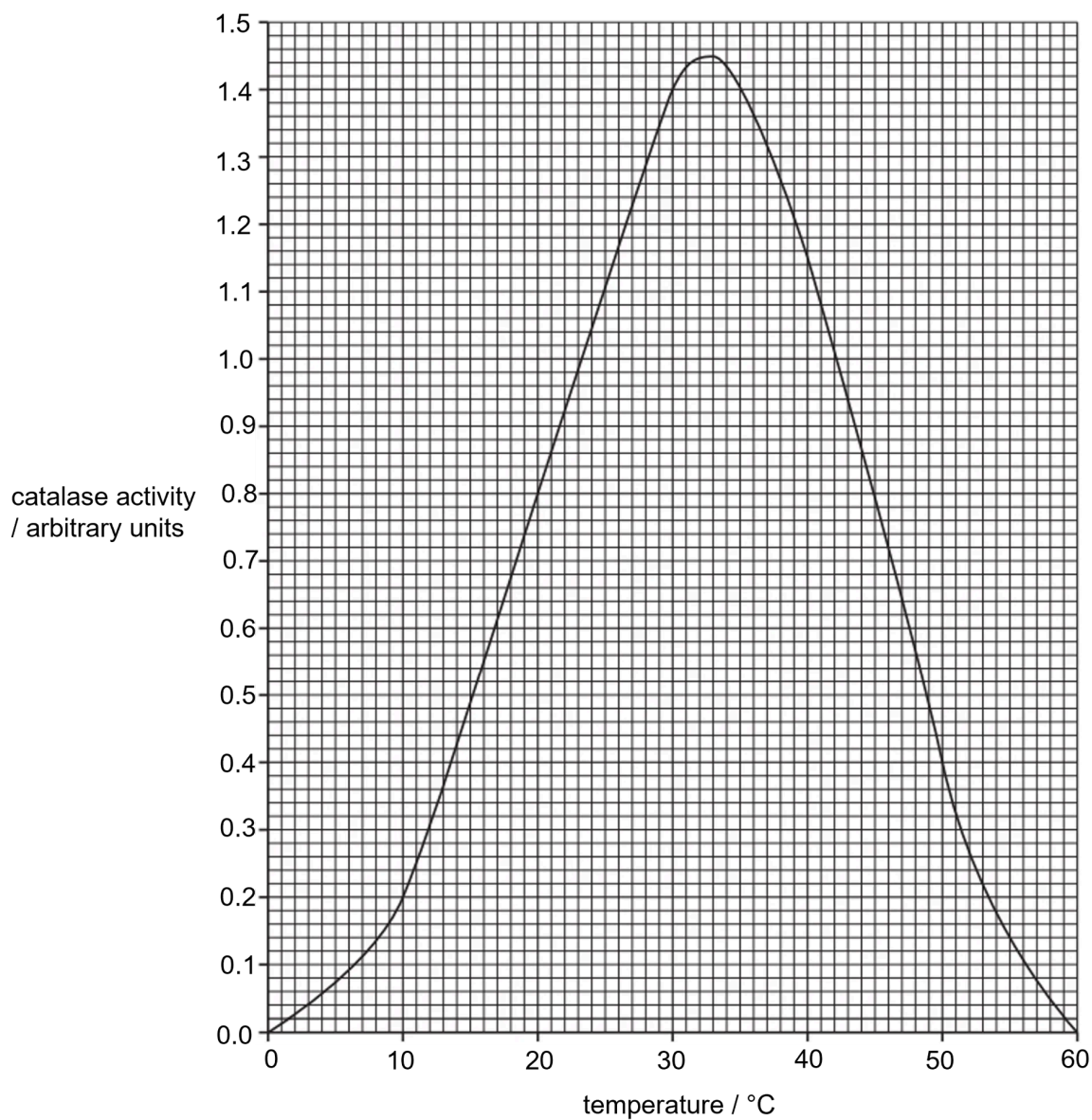


Fig. 1

(i)

State the temperature at which catalase is most active in Fig. 1. Give your answer in °C.

[1]

(ii)

Explain why there is no enzyme activity at 60°C.

[1]

.....
..... (2 marks)

(b) State **one** factor other than temperature that affects enzyme activity.

..... (1 mark)

(c) Digestive enzymes are used in chemical digestion to break down large insoluble molecules to smaller soluble molecules.

The boxes show some large insoluble molecules, some digestive enzymes and some smaller soluble molecules that are produced during digestion.

Draw **one** straight line from each enzyme to the insoluble molecule it acts on.

Draw another line from each enzyme to the smaller soluble molecule that is produced.

Draw a total of **six** lines.

insoluble molecule

fat

enzyme

amylase

soluble molecules

amino acids

protein

lipase

fatty acids
and glycerol

starch

protease

sugars

[5]

(5 marks)

(d) Digestive enzymes are secreted in different parts of the body.

Gastric proteases are found in the stomach.

Suggest the optimum pH for gastric protease.

..... **(1 mark)**

7 Some terms used to describe enzyme-controlled reactions are listed.

- 1 catalyst
- 2 product
- 3 protein
- 4 substrate

Which terms describe an enzyme?

- A.** 1 and 3
- B.** 1 and 4
- C.** 2 and 3
- D.** 2 and 4

(1 mark)

8 One of the enzymes produced by this cell is lipase.

Describe how lipase functions in the digestive system, including the products formed.

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(3 marks)